

From Ground Station to Flying 5G Relay

Progress on a Drone-Oriented 5G Testbed

Research Progress Recap

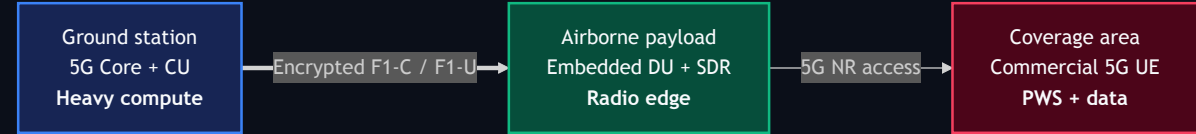
Presenter: Research Team

The idea in one picture

Project Concept

Instead of carrying an entire heavy 5G base station on a drone, we **separate the lightweight radio** parts from the heavy computing intelligence on the ground.

The core network brain stays safe on the ground, while the drone serves as a lightweight flying relay.



Why this project matters

Disaster & Emergency Coverage

Temporary coverage is vital after **floods, forest fires, or critical outages** where ground infrastructure is destroyed.

A drone can **restore emergency coverage within minutes**. The challenge is keeping airborne equipment light and power-efficient.



What CU/DU split means

Architecture Concept

We divide a 5G base station into two parts:

- **Core Network & CU:** The ground-based "brain" managing user sessions, security, and protocols.
- **DU & Radio antenna:** The drone-carried "relay" transmitting actual high-speed radio waves.
- **Backhaul Link (F1):** Wireless connection linking them.

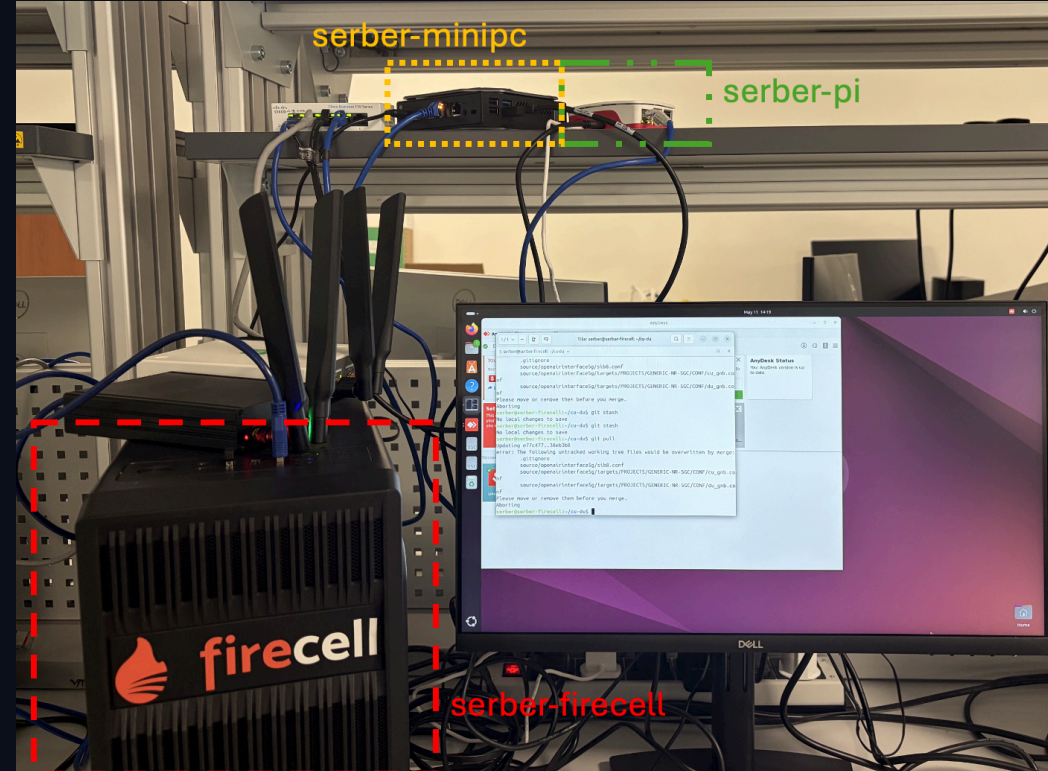


The physical testbed

Hardware Setup

We validated a real-world benchtop setup using accessible hardware:

- **serber-firecell**: Ground Core & CU.
- **serber-minipc**: Remote DU unit.
- **Raspberry Pi 5**: Portable DU candidate.
- **USRP B210**: Software-programmable antenna.
- **Quectel Modem**: Wireless backhaul link.

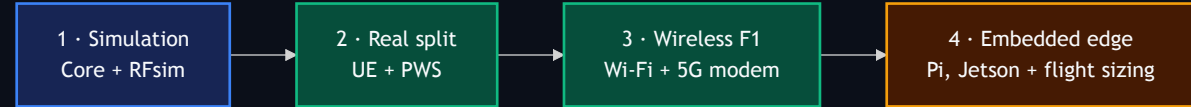


Progress timeline

Milestones Timeline

The system progressed systematically from software simulations to real-world wireless hardware loops.

Key achievements include connecting a **commercial phone**, broadcasting **emergency warnings (PWS)**, and implementing **5G WireGuard backhauling**.



The main engineering pivots

Hurdles & Adaptations

Faced with real-world hardware limits, the project shifted direction to find viable solutions:

We resolved **protocol blocks**, **hardware overloading**, and **signal saturation** to reach a stable prototype.

1. Jetson Kernel and I/O

A custom SCTP kernel plus CPU, IRQ, and USB tuning made the embedded DU viable.

2. Single-Radio Overload

One radio could not serve both access and backhaul. Dedicated B210 to phone, and Quectel modem to backhaul.

3. Transport and Scheduler

MTU/MSS and BLER-threshold tuning removed the early 23 Mbps split ceiling.

What works today

Current Status

Our current system has successfully validated all key features of the split 5G architecture.

The remaining work is to turn best-observed runs into a controlled, publishable measurement campaign and complete the flight integration.

- **Commercial phone attached**
- **Public Warning System SIB8**
- **CU/DU split F1 separation**
- **Wireless F1 tunnel over WiFi/5G**
- **Raspberry Pi 5 DU timing fix**
- **End-to-end backhaul speed**
- **Split scheduler bottleneck fix**
- **Repeated benchmark campaign**
- **Flight integration**

Done

Done

Done

Done

Done

Done

Done

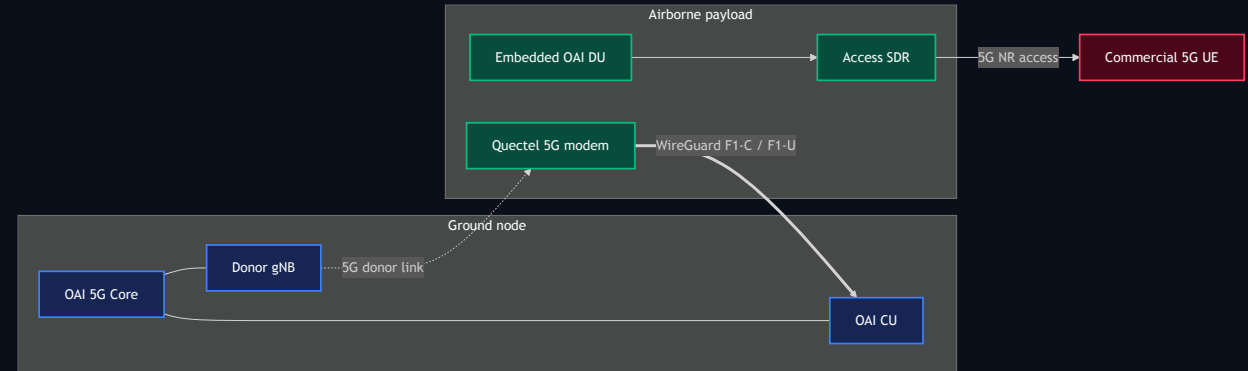
Pending

Pending

Current target architecture

System Layout

- **Ground:** 5G core, CU, and donor cell.
- **Payload:** Embedded DU, access SDR, and Quectel modem.
- **Transport:** WireGuard carries F1-C and F1-U over the donor 5G link.



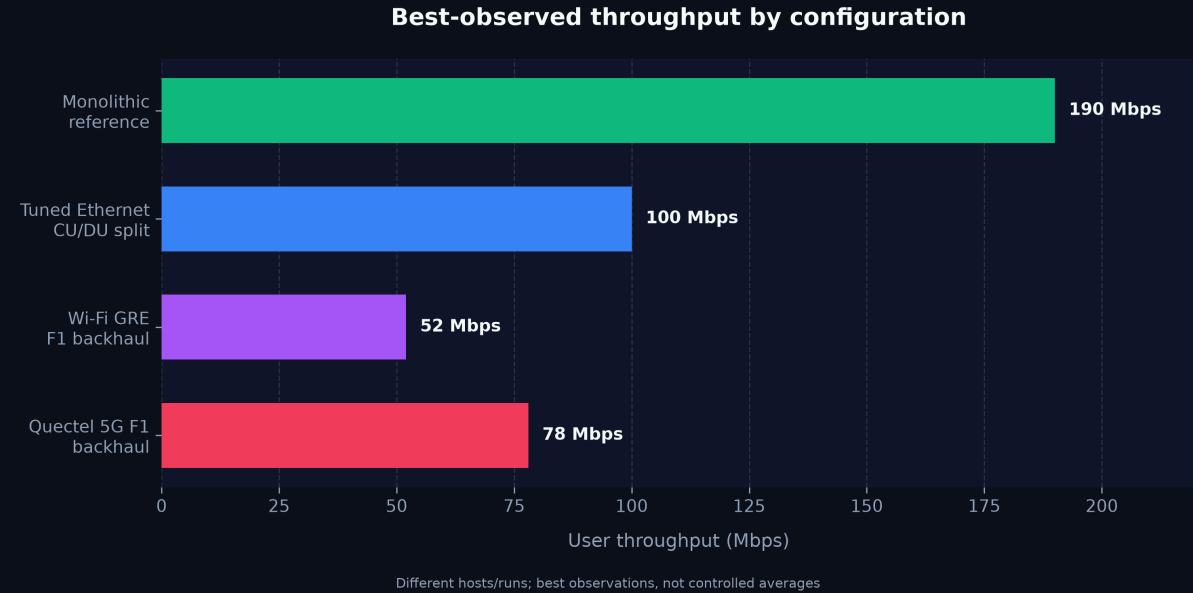
Tuning removed the early 23 Mbps ceiling

Best-observed throughput

Later tests overturned the early split ceiling:

- **100 Mbps** tuned Ethernet
- **52 Mbps** Wi-Fi/GRE
- **78 Mbps** Quectel/WireGuard
- **190 Mbps** monolithic peak

Interpretation: MTU/MSS behavior and BLER thresholds—not the split alone—were the recoverable constraints. Values are best observations.



What is next

Roadmap to a defensible flight artifact

The separated 5G network and wireless F1 paths now work. The next phase turns best-observed demonstrations into a repeatable, flight-safe research artifact.

1. **Freeze the stack:** Publish the exact OAI revision, patches, and configs.
2. **Repeat the campaign:** Run controlled trials with synchronized metrics.
3. **Publish evidence:** Release sanitized samples and packet-path proof.
4. **Validate the light radio:** Qualify B205mini-i before choosing the small-drone path.
5. **Engineer the flight:** Bench-test power, cooling, RF isolation, mounting, and failsafes.